STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 13: Oscillations

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

Define period. [1]
Frequency-period relation? [1]
Restoring force in SHM? [1]
Where is speed maximum in SHM? [1]
Where is acceleration magnitude maximum? [1]
Spring angular frequency? [1]
Small-angle pendulum period? [1]
Total spring energy for amplitude A? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

Period .5s: frequency? [2]
m=.5kg,k=200N/m: period? [2]
For x=Acosomegat, acceleration? [2]
Pendulum length becomes 4l: period factor? [2]
Velocity at an extreme? [2]
Condition for pendulum SHM approximation? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

m=.5,k=200: angular frequency and period? [3]
A=.1m,omega=4: maximum speed and acceleration? [3]
Pendulum length 1m,g=9.8: period? [3]
Describe energy in one SHM cycle. [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

k=50N/m,A=.2m: total energy and U at x=.1m? [4]
Why does a real oscillator stop? [4]

End of paper - check working, units, signs and significant figures.